

Phase 7c — Universality Breadth Brief

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2026-04-25

1 Purpose

Phases 6 and 7 demonstrated that two new spectral-entropy variants — `scalar_entropy_normalized` (Path B) and `entropy_target H*=0.5` (Path A) — universally Pareto-dominate the original `scalar_entropy` regulariser across four stress datasets (spirals, circles, MNIST plain MLP, CapsMLP MNIST). Phase 7c extends this stress test to six additional dataset/architecture combinations that were tested only with the original `scalar_entropy` in phases 1–5:

- Deep skip-connection architectures: ResMLP-20, HighwayMLP-20
- MNIST-shape image siblings: FashionMNIST, KMNIST
- Different output-class structure: EMNIST Letters (26 classes)
- Different input shape: SVHN (32×32 colour)

If both new arms preserve sign and magnitude (or amplify) on these six combinations, the universality claim covers **5 architectures** × **9 datasets** with no significant negative outliers.

Auto-fires after Views PH7b suite finished marker. Estimated wall-clock: ~17 hours. Output CSVs land in `data/benchmarks/thesis_iv_hard_2026042?_*.csv`.

2 Test arms

Both arms target `--view dataflow` and `--target-entropy 0.5`. Original `scalar_entropy` arm runs alongside as the paired baseline.

Arm	Formula	Hyperparameters
<code>scalar_entropy_normalized</code>	$\lambda \cdot H(A) / \log_2 \text{rank}(A)$	none beyond λ
<code>entropy_target</code>	$\lambda \cdot (H_{\text{norm}} - H^*)^2$	$H^* = 0.5$

Code references in `python/benches/thesis_iv_hard/run_benchmark.py`: the `scalar_entropy_normalized` branch is at lines 1380–1397, the `entropy_target` branch at 1398–1416.

3 Experiment cards

3.1 ResMLP-20 — deep residual MLP

Dataset / arch: MNIST handwritten digits (60k train, 10k test); 20 residual blocks of width 16 (`mnist_resnet_20`). **Config:** 33 seeds × 5 epochs, $\lambda=0.1$, reg-every-n=10, dataflow.

Original (phase 1, scalar_entropy): $\Delta=+0.046$ pp, $t=+1.76$ (marginal); replicated $\Delta=+0.070$ pp, $t=+1.89$.

Expected outcome: both new arms should keep the marginal positive but tighten the t-statistic. Deeper architectures should benefit *more* from normalisation than plain MLPs because their effective rank grows with depth.

```
python3 python/benches/thesis_iv_hard/run_benchmark.py \
  --datasets mnist_resnet_20 \
  --arms baseline scalar_entropy_normalized \
  --seeds 33 --epochs 5 --lam 0.1 --reg-every-n 10 \
  --view dataflow --target-entropy 0.5

python3 python/benches/thesis_iv_hard/run_benchmark.py \
  --datasets mnist_resnet_20 \
  --arms baseline entropy_target \
  --seeds 33 --epochs 5 --lam 0.1 --reg-every-n 10 \
  --view dataflow --target-entropy 0.5
```

3.2 FashionMNIST — image sibling

Dataset / arch: Fashion items (28×28 grayscale, 10 classes); plain MLP $784 \rightarrow 16 \rightarrow 8 \rightarrow 10$ (`fashion_mnist`).
Config: 33×15 , $\lambda=0.1$.

Original: $\Delta=+0.039$ pp, $t=+1.13$ (null).

Expected outcome: sibling test. If MNIST’s anchor result was MNIST-specific, the original null persists; if normalisation is the real fix, both new arms convert FashionMNIST into a positive of similar magnitude ($\approx +0.10$ pp expected).

```
python3 python/benches/thesis_iv_hard/run_benchmark.py \
  --datasets fashion_mnist \
  --arms baseline scalar_entropy_normalized \
  --seeds 33 --epochs 15 --lam 0.1 --reg-every-n 10 \
  --view dataflow --target-entropy 0.5

python3 python/benches/thesis_iv_hard/run_benchmark.py \
  --datasets fashion_mnist \
  --arms baseline entropy_target \
  --seeds 33 --epochs 15 --lam 0.1 --reg-every-n 10 \
  --view dataflow --target-entropy 0.5
```

3.3 KMNIST — Japanese hiragana

Dataset / arch: 70k handwritten hiragana (28×28 grayscale, 10 classes); plain MLP (`kmnist`). Backed by the HuggingFace `tanganke/kmnist` parquet mirror (the official `codh.rois.ac.jp` torchvision URL is dead). Decoded arrays cached to `datasets/kmnist/kmnist_decoded.npz`. **Config:** 33×15 , $\lambda=0.1$.

Original: $\Delta=-0.029$ pp, $t=-0.33$ (null).

Expected outcome: different script family from MNIST. If the new arms produce a positive, universality covers cross-script generalisation.

```
python3 python/benches/thesis_iv_hard/run_benchmark.py \
  --datasets kmnist \
  --arms baseline scalar_entropy_normalized \
  --seeds 33 --epochs 15 --lam 0.1 --reg-every-n 10 \
  --view dataflow --target-entropy 0.5

python3 python/benches/thesis_iv_hard/run_benchmark.py \
  --datasets kmnist \
```

```
--arms baseline entropy_target \
--seeds 33 --epochs 15 --lam 0.1 --reg-every-n 10 \
--view dataflow --target-entropy 0.5
```

3.4 SVHN — Street View House Numbers

Dataset / arch: 73k digit crops ($32 \times 32 \times 3$ colour, 10 classes); SVHNNet $3072 \rightarrow 64 \rightarrow 32 \rightarrow 10$ with the $3072 \rightarrow 64$ input projection skipped from the regulariser (`svhn`). **Config:** 15×20 , $\lambda=0.1$.

Original: $\Delta = -0.137$ pp, $t = -0.55$ (null, directionally negative).

Expected outcome: hardest test in the batch — colour input, small seed count. If both new arms shift SVHN toward zero or positive, the failure-mode-mitigation claim is supported on 3-channel inputs.

```
python3 python/benches/thesis_iv_hard/run_benchmark.py \
--datasets svhn \
--arms baseline scalar_entropy_normalized \
--seeds 15 --epochs 20 --lam 0.1 --reg-every-n 10 \
--view dataflow --target-entropy 0.5

python3 python/benches/thesis_iv_hard/run_benchmark.py \
--datasets svhn \
--arms baseline entropy_target \
--seeds 15 --epochs 20 --lam 0.1 --reg-every-n 10 \
--view dataflow --target-entropy 0.5
```

3.5 EMNIST Letters — 26-class MNIST sibling

Dataset / arch: 145k handwritten letters (28×28 , 26 classes, labels shifted to $0 \dots 25$ from torchvision's $1 \dots 26$ convention); MNISTNetSmallNClass(26) (`emnist_letters`). **Config:** 33×15 , $\lambda=0.1$.

Original: $\Delta = +0.170$ pp, $t = +1.24$ (directional, not significant). **Note:** σ drops from 0.941 to 0.684 (27% variance reduction — strongest variance-drop of any image dataset under `scalar_entropy`).

Expected outcome: this is the most likely big-win conversion. Underlying variance-reduction signal is already strong; both new arms should bring the t-stat into significance ($p < 0.05$) once normalisation handles the larger 26-class output dimension better.

```
python3 python/benches/thesis_iv_hard/run_benchmark.py \
--datasets mnist_letters \
--arms baseline scalar_entropy_normalized \
--seeds 33 --epochs 15 --lam 0.1 --reg-every-n 10 \
--view dataflow --target-entropy 0.5

python3 python/benches/thesis_iv_hard/run_benchmark.py \
--datasets mnist_letters \
--arms baseline entropy_target \
--seeds 33 --epochs 15 --lam 0.1 --reg-every-n 10 \
--view dataflow --target-entropy 0.5
```

3.6 MNIST HighwayMLP-20 — deep gated MLP

Dataset / arch: MNIST; 20-block Highway MLP (gated skip connections at width 16) (`mnist_highway_20`). **Config:** 33×15 , $\lambda=0.1$.

Original: baseline was significantly positive in phase 4; single-line aggregate is in `RESULTS_VIEWS_SUITE.md`. Expectation in the same neighbourhood as ResMLP-20.

Expected outcome: Highway gating differs from ResMLP’s additive skip; the question is whether normalisation + target entropy still land positive across both deep-MLP architectures, confirming the claim is architecture-independent among skip-connection styles.

```
python3 python/benches/thesis_iv_hard/run_benchmark.py \
  --datasets mnist_highway_20 \
  --arms baseline scalar_entropy_normalized \
  --seeds 33 --epochs 15 --lam 0.1 --reg-every-n 10 \
  --view dataflow --target-entropy 0.5

python3 python/benches/thesis_iv_hard/run_benchmark.py \
  --datasets mnist_highway_20 \
  --arms baseline entropy_target \
  --seeds 33 --epochs 15 --lam 0.1 --reg-every-n 10 \
  --view dataflow --target-entropy 0.5
```

4 Aggregated expected outcomes

If the universality claim holds across phase 7c:

Dataset	original Δ pp	expected B Δ	expected A Δ
ResMLP-20 (MNIST)	+0.046 marginal	+0.05 to +0.10 sig	+0.05 to +0.10 sig
FashionMNIST	+0.039 null	+0.08 to +0.12 sig	+0.08 to +0.12 sig
KMNIST	−0.029 null	−0.02 to +0.03 null	0 to +0.05 null
SVHN	−0.137 null	−0.05 to +0.05 null	−0.05 to +0.05 null
EMNIST Letters	+0.170 direct.	+0.20 to +0.25 sig	+0.20 to +0.25 sig
HighwayMLP-20	(re-test)	+0.04 to +0.10 marg./sig	+0.04 to +0.10 marg./sig

If observed Δ falls outside these ranges by more than 50%, the claim needs qualification per dataset.

5 After phase 7c

Re-run `python3 python/benches/thesis_iv_hard/aggregate_views_suite.py` to refresh `RESULTS_VIEWS_SUITE` with the 12 new rows. Then regenerate the LaTeX report (`reports/thesis_iv_views_suite.tex`) for a final 132-experiment summary.